



Liquor store in Madrid, Spain financial analysis

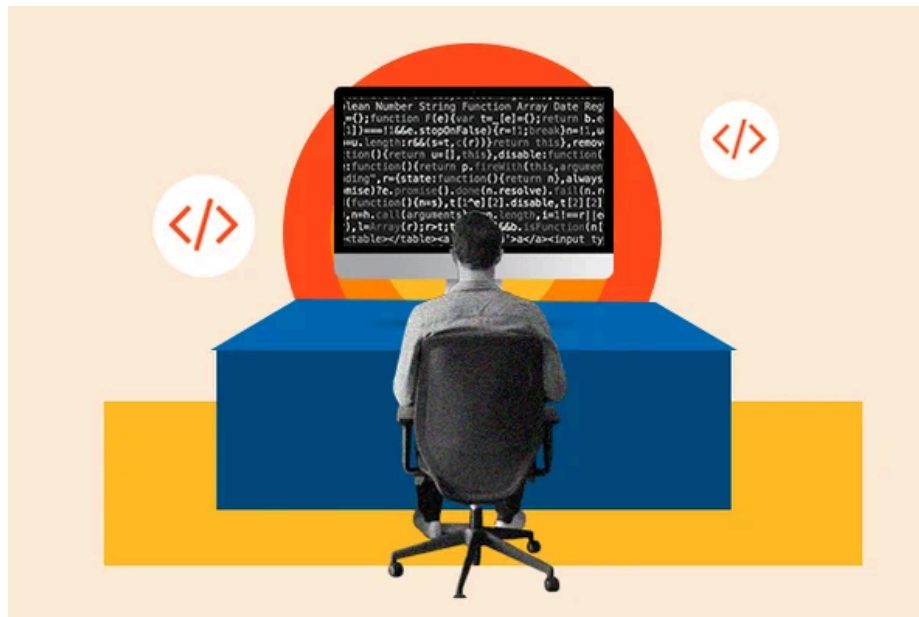
Made by:

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Introduction:

In 2012 Harvard Business Review published an article, where it indicated that **Data Science would be the sexiest profession of the 21st century. [1]**

Is it really possible that, in today's digital era, some businesses are still not leveraging up-to-date analytical tools? **What about Python?**



In this essay, the application of Python for descriptive analytics in a Madrid-based liquor store and other analytical tools will be explored. Along with the analysis that will be provided, sales data will be analyzed, a try to spot common patterns/trends will be made and a valuable proposal of actions that will detonate the analysis based on multiple questions will be provided.

With the already mentioned approach, improvement areas will be covered, which address the current situation of the store, providing an analysis for a good understanding of the situation and exponential growth strategies.

Descriptive statistics:

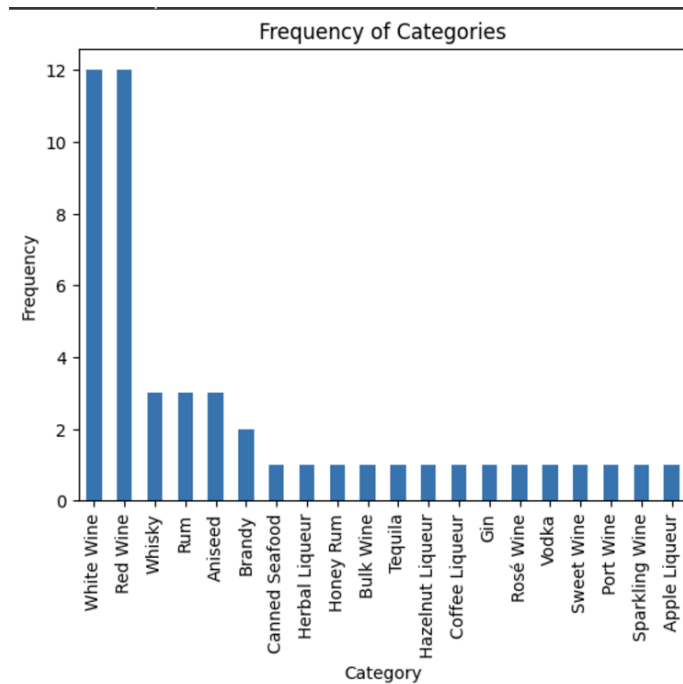
	Unidades	€/UD.	TOTAL	Discount	Net Total (€)
count	49.000000	49.000000	49.000000	49.000000	49.000000
mean	5.755102	8.331375	33.927551	0.048571	31.209327
std	7.022377	6.766385	37.744996	0.038837	33.381944
min	1.000000	1.550000	4.000000	0.000000	4.000000
25%	1.000000	4.110000	9.350000	0.000000	9.350000
50%	3.000000	6.250000	18.750000	0.050000	17.812500
75%	6.000000	10.500000	39.750000	0.070000	37.762500
max	36.000000	31.000000	159.000000	0.150000	143.100000

Questions that detonate the analysis:

A set of questions will follow, which will help us detonate the analysis that will be provided. As well, as mentioned in the rubric, at least one chart, one for each chart will be included for each triggering question:

What are the most frequent product categories across the 49 entries we got from the store located in Madrid, Spain?

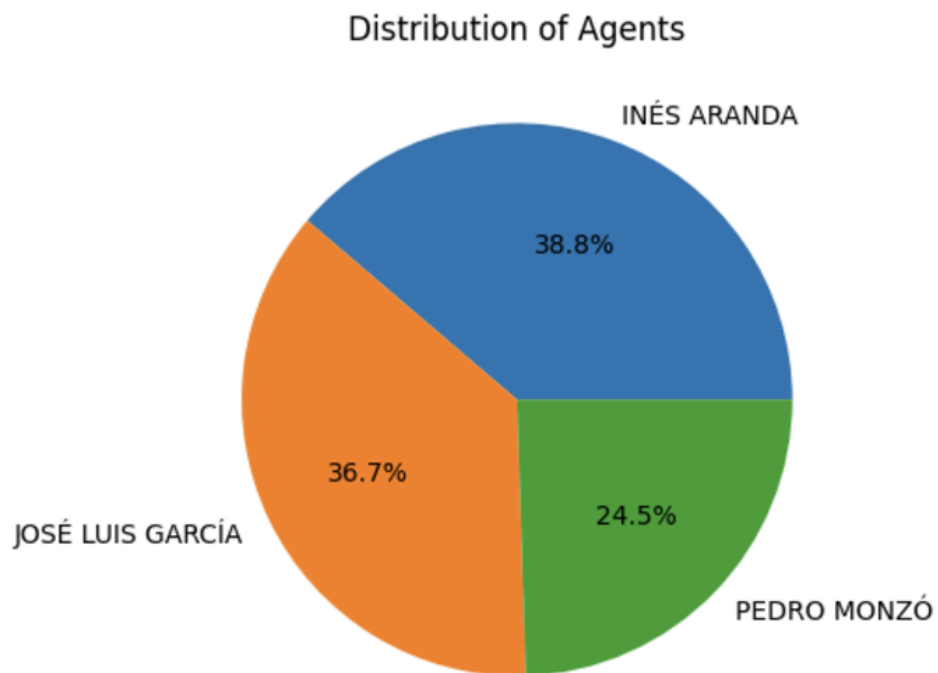
In the following graph we can see all the product categories in a bar graph:



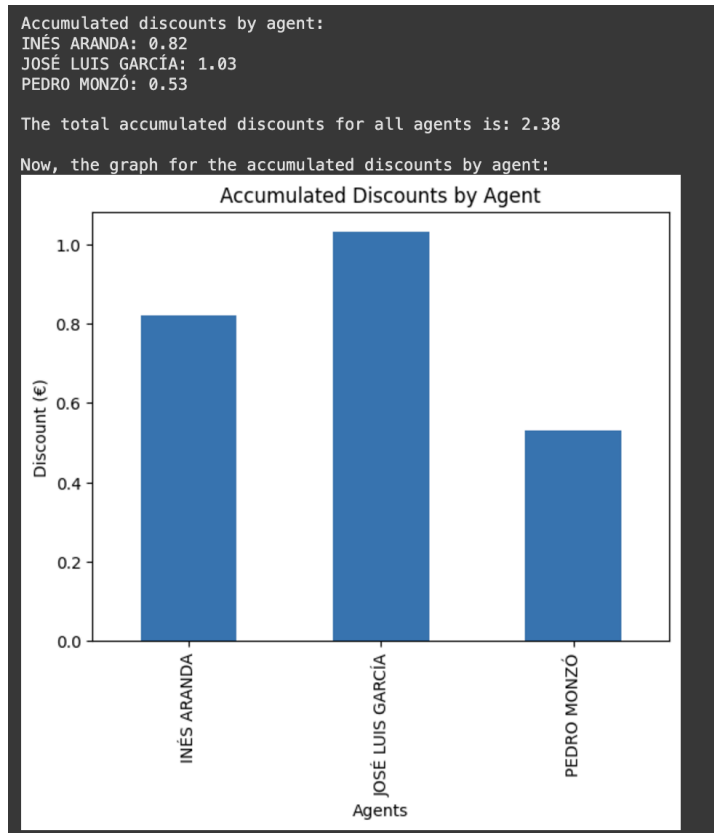
This is key information that is not readily available to the naked-eye, since when just taking a quick look at the Excel, all product categories are equally the same, but if we ask Python to graph this qualitative information, as in this case, we can see that **Red and White wine** sell equally the same (**a total of 12 times in both for just 49 entries!**) and that all the other products follow a similar-looking pattern. This information provided will help us a lot when developing strategic planning in order to address this situation.

What's the agent that purchases products more often? Does he get the most discounts in comparison with the other agents?

As it can be observed in this pie chart, Inés Aranda is the agent that makes up to 38.8% of the total purchases of the liquor store:



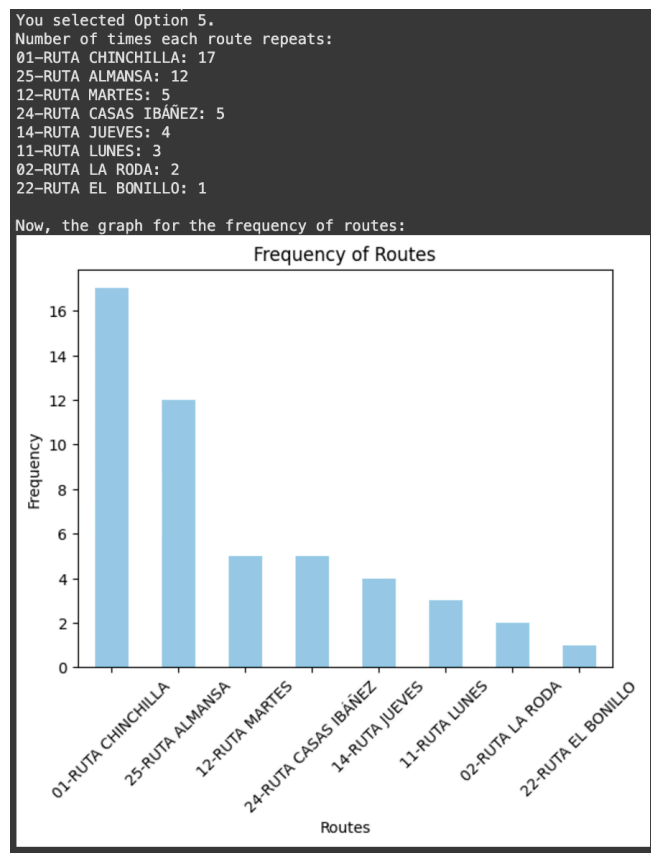
If we continue with the analysis, and now get what was the accumulated discounts each of the three agents got, see what was the total discounts that the liquor store gave in total for 49 entries and observe a bar graph that shows this information, we get the following:



What was the most repeated route? If we visualize a graph, do we see a visible pattern that gives us valuable insights?

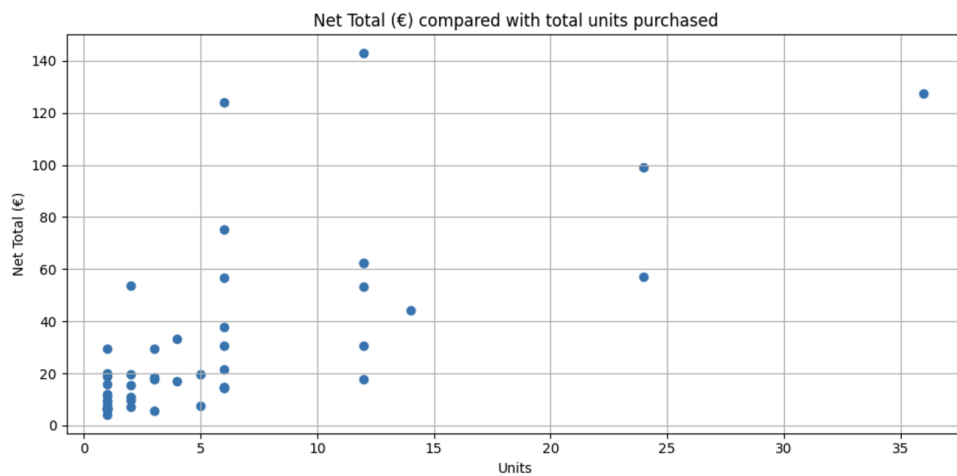
The most repeated route without a doubt was: **01-RUTA CHINCHILLA**: a total of **17** times.

We can see all the repeated routes and a bar graph with much more clarity with the next code:

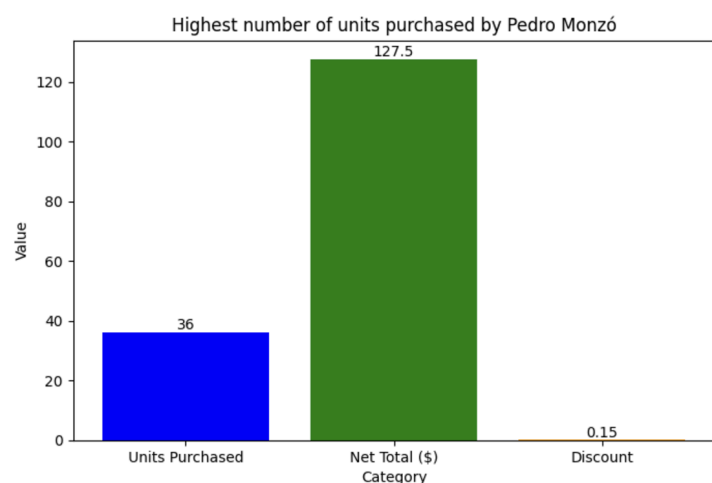


What's the relationship between unities sold and the Net Total in euros (€)?

We can see from the following scatter plot that the highest net total that was purchased during this period of time and the 49 registered entries was of: \$143.1, representing 12 units.



This is quite interesting, since the highest number of units purchased by a customer was of **36**, its net total was of **\$127.5**, getting the highest registered discount: **0.15** and purchased by **Pedro Monzó**, here we can see the following graph made in Python that represents the mentioned information:



My proposal of actions that add value to the strategies that the business must take, this according to the problems raised:

- **Most frequent product categories:**

After evaluating that a total of 12 times (24.5% of 49 total entries) were based on just red and white wine, and the categories that ranked among the lowest were whisky, brandy, cider, etc, I would propose the liquor store to increase their inventory levels of **White and Red wine**, so they never run out of them, and specially reducing their stock for products that have way less frequency.

As well, it would be great to make bundle-promotions that give the client a discount when they purchase red and white wine bottles together.

This can boost the interest of the agents, since these are the two types of wine that rank among the highest in frequency.

**For this case it's not possible to change their already existing Excel sheet, since the information provided from the liquor store does not have any relation with the stock the store is handling, we just got the information from their sales.*

- **Frequency of agents and accumulated discounts:**

After analyzing that Inés Aranda accounts for 38.8% of purchases, followed closely by José Luis García (36.7%) and Pedro Monzó (24.5%). It was interesting to see that José Luis García was the one that received the highest percentage of discounts (1.03) and that Pedro Monzó gets the least (0.53)

Some of the suggestions that can be made from this valuable information are:

Loyalty programs can be implemented to agents that buy the highest quantity of products from them, like in this case Inés Aranda and José Luis García, by offering them exclusive discounts to improve the loyalty of theirs.

In this case, it is advisable to offer 2-5% loyalty discounts on future purchases per agent. After doing so, analyze if the consumer behavior from the agents changes and record the results from before, during and after making use of this strategy.

- **Frequency of routes:**

As already mentioned, Ruta Chinchilla was the most frequent one (a total of 17 times), followed by Ruta Almansa (with a total of 12 times), taking into account that the other routes were not that common (they occurred just one time). This valuable information gives us the opportunity to propose actions that can help the liquor store to focus on the routes that can give them the best selling opportunities:

If they focus their logistics attention on Ruta Chinchilla and Ruta Almansa they can increase their delivery frequency for them to improve their customer satisfaction, and, at the same time, investigating growth opportunities in less frequented routes so they can achieve a balance in delivery, so they can make use of their resources more efficiently.

Relationship between units sold and net total:

Finally, as already mentioned, the highest units purchased per agent was of 36, which corresponded to a grand total of €127.5, made by Pedro Monzó (with a discount of 0.15).

The scatter plot shows a relationship, in which the higher the units sold, generally speaking, the higher the net totals (meaning that not many discounts are not based only on the accumulated money that the agents are to pay, but as well on the category of product they're purchasing).

With these relevant insights it can be proposed for the liquor store to offer more discounts for bulk purchases, so the higher the sale, the greater the discount.

Conclusion:

In conclusion, and taking into account the problems that I had to go through when performing this data analysis and developing the Python code were, at first, **finding the actual database**. After getting it, another problem then was: ***What type of questions are we supposed to ask? What procedure should we follow?*** After making the proposals and starting with the code everything was much smoother once we actually knew what information we were expecting the code to provide us, to which the graphs and plots followed to conclude the analysis.

All this wouldn't have been possible without the use of machine learning and statistical tools that are to be found in Python. All the information that was covered during this course was crucial in order to actually capture the analyses in the way we wanted and customized them to the specific needs of our database, something that when just using Excel, a great tool for performing data analysis and plotting graphs, even though it's much more easy it's far less customizable. Therefore, the course was undoubtedly relevant and therefore the materials that were covered were the key to success to solve the problems we at first detected: **What strategies can we propose the liquor store after having done the analysis and obtained relevant insights?**

Sources

[1] Data Scientist: The Sexiest Job of the 21st Century. (2024, March 19). Harvard Business Review. <https://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century>

[2] Plot types — Matplotlib 3.9.2 documentation. (n.d.).
https://matplotlib.org/stable/plot_types/index.html

[3] Vishal. (2022, February 16). Generate random strings and passwords in Python. PYNative. <https://pynative.com/python-generate-random-string/>